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ORIGINAL PAPER

Vortex ring state as a cause of a single-rotor helicopter unanticipated yaw

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Abstract
Single-rotor helicopters under certain conditions can enter the unanticipated yaw rotation, which often leads to the aviation accident. However, the causes of this flight mode have not been definitely determined yet. Typically, the main researcher's proposal is that unanticipated yaw occurs, because helicopter tail rotor enters the vortex ring state. Authors of this paper decided to study the vortex ring state from the point of propeller thrust variation while being airflow blown. Experiments were carried out with a propeller in a wind tunnel, in which they varied the airflow velocity and angle between propeller axis of rotation and airflow velocity vector to imitate tail rotor side-blowing, while varying helicopter sideslip angle. Experiments results showed that at the vortex ring state there was a tail rotor thrust drop but it was not significant and thrust restored when the angle between propeller axis of rotation and airflow velocity vector exceeded 40°. But when propeller was axial blown in the direction against propeller thrust vector with airflow velocity 10 m/s the propeller thrust dropped by 80% relative to the thrust without blowing, which may lead to the occurrence of unanticipated yaw rotation. This suggests that helicopter wind blowing against tail rotor thrust vector is more dangerous than blowing in the opposite direction which leads to the vortex ring state.

Keywords Helicopter · Unanticipated yaw · Loss of tail rotor effectiveness · Vortex ring state · Propeller · Wind tunnel

1 Introduction

Practice of single-rotor helicopters operation reveals that under certain conditions they can enter the unanticipated (unexpected for the pilot) yaw rotation. Prevention of entering this flight mode and timely rotation stop in case of its occurrence depend, among other things, on timely and adequate pilot actions. It is important to understand under what conditions an unanticipated yaw occurs and what is its cause. However, this issue has not been finally resolved yet. Helicopter unanticipated yaw may be due to the loss of effectiveness of tail rotor (TR) which is designed to compensate the main rotor (MR) torque and for helicopter yaw control. Loss of tail rotor effectiveness (LTE) can be caused by failures of helicopter elements and systems as well as external conditions including wind. Failures and external conditions that lead to structural defects are not considered in this paper since their impact on TR operation is obvious. The situation is more complicated with aerodynamic effects that can have a negative impact on TR and helicopter integrally. Literature analysis on the unanticipated yaw of a single-rotor helicopter suggests that there are three main reasons for this phenomenon:

- TR thrust reduction because of MR and TR interference.
- Aerodynamic yaw moment occurrence when blowing the helicopter from behind because of its directional stability.
- TR thrust reduction because of the vortex ring state.

In most papers dedicated to this phenomenon [1–12], it is considered that the main reason that triggers unanticipated yaw is the well-known vortex ring state which the TR falls into when it is blown by a side wind or when it turns sharply on the yaw. Vortex ring state occurs when TR is blown by airflow against the throwback flow (flow ejected by TR). If the airflow velocity is equal in value and opposite in direction to the TR throwback velocity (or ejection velocity), then the

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Fig. 1 Tested propeller



Table 1 Propeller geometric characteristics

Blade section number	Blade section radius, mm	Blade section chord, mm	Airfoil thickness, mm	Blade section pitch angle, °
1	10	13.0	5.5	28
2	20	14.5	4.5	30
3	30	18.0	4.0	31
4	40	20.0	3.2	24
5	50	21.0	3.1	22
6	60	20.0	3.0	18
7	70	18.0	2.8	17
8	80	13.5	2.1	16
9	90	8.0	1.2	9

sucked air will not be ejected by TR but sucked again so there will be a circulating air movement—the vortex ring state [13–18]. According to the propeller impulse theory, if propeller does not eject airflow, then there should be no propeller thrust. As a result, TR loses its efficiency: it cannot compensate the MR torque and unanticipated yaw occurs. However, some papers have appeared recently [19] in which LTE while TR is being blown by the airflow is doubted. Some flight experiments with real helicopters [20, 21] also do not confirm the idea of LTE because of airflow blowing which leads to the occurrence of vortex ring state. Due to the presence of such conflicting conclusions about vortex ring state and its role in unanticipated yaw rotation phenomenon authors of this paper decided to study the vortex ring state concerning propeller thrust variation while being blown by airflow. Similar experiments with propeller axial blowing have been conducted earlier [22], however, authors of this paper know nothing about experimental research of vortex ring state with airflow side-blowing. Meanwhile, in case of helicopter, yaw rotation the angle between TR axis of rotation and airflow velocity vector varies.

2 Experimental setup and research method description

At this initial stage of research, the fixed-pitch propeller with blades geometric twist was chosen as a research object. It is shown in Fig. 1.

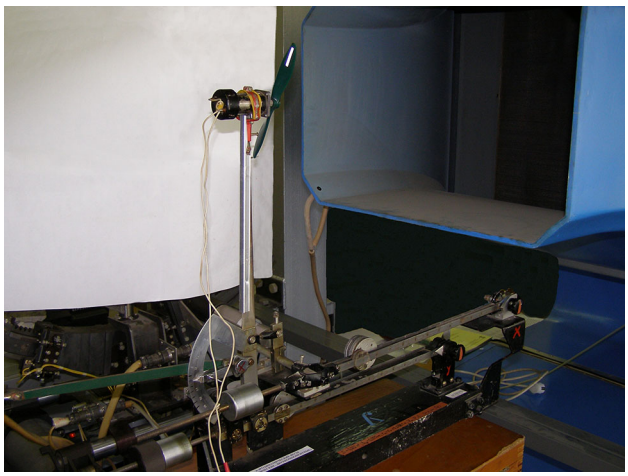


Fig. 2 Experimental setup in the wind tunnel test chamber

Propeller radius is 90 mm, propeller pitch is 125.5 mm. Detailed propeller geometric characteristics are presented in Table 1.

Propeller was mounted on the electric motor attached to the mechanical scales by which the propeller generated thrust was measured.

This experimental setup was placed in the wind tunnel test chamber of Moscow State Technical University of Civil Aviation Aerodynamics and Aircraft Structures Department (Fig. 2).

Wind tunnel was generating airflow blowing on the propeller. Scales were mounted on the rotatable base that could

allow varying the angle β between propeller axis of rotation and airflow velocity vector. It could give an opportunity to imitate tail rotor side-blowing while varying helicopter sideslip angle.

During the experiment, the angle between propeller axis of rotation and airflow velocity vector varied from 0 to 90 degrees with a step of 10°. Airflow velocity V_0 generated by wind tunnel varied from 2 to 10 m/s with a step of 1 m/s. Propeller was blown for increasing blades angles of attack (against the direction of induced velocity, $V < 0$) and for decreasing blades angles of attack (in the direction of induced velocity, $V > 0$).

Electric motor input power was 16.1 W, in these conditions, propeller rotation speed varied from 5410 to 6280 rpm because of airflow affecting. Propeller rotation speed was measured by the laser digital tachometer.

Propeller thrust without airflow blowing was 1.12 N with rotation speed of 5517 rpm. Thrust value was defined by extracting electric motor drag from the force measured by mechanical scales. Propeller throwback velocity was about 6 m/s, it was measured by the anemometer.

Wind tunnel experiment was carried out as follows. Airflow velocity was set, electric motor was switched on, propeller gained speed, propeller thrust and rotation speed were measured. Then, the angle between propeller axis of rotation and airflow velocity vector was varied by rotating the scales base. Propeller thrust and rotation speed were measured again. Results of measurements are presented and analyzed below.

3 Experiments results and analysis

Figure 3 below shows graphs of dependencies of propeller thrust T on airflow velocity V at different angles β between propeller axis of rotation and airflow velocity vector.

It can be seen from the presented graphs that at low angles β and positive airflow velocity the propeller thrust decreases while velocity increases. This is due to propeller blades sections angles of attack decrease. At negative airflow velocity, the propeller thrust increases while velocity absolute value increases. This is due to propeller blades sections angles of attack increase. The general character of the graph at $\beta = 0$ corresponds to the graphs given in [22] (Fig. 4). In these graphs, the dependencies of relative thrust c_T on relative airflow velocity V_0 at different propeller pitch angles φ are provided. However, in Fig. 3, in the certain range of negative velocities, there is a drop of thrust with a minimum at about $V = 6$ m/s. This velocity corresponds to the propeller throwback velocity measured earlier. In these blowing conditions, the vortex ring state forms. Therefore, there is a thrust drop at the vortex ring state. Nevertheless as seen from the graphs, the propeller thrust does not drop to zero in these conditions, although according to earlier mentioned impulse theory there should be no propeller thrust. This conflict statement was mentioned by the author of the propeller impulse theory Yuriev as he noted that the impulse theory is not applicable to the vortex ring state [22]. Meanwhile, there is no thrust drop on the graphs provided in his work at all (Fig. 4). It seems that the presence or absence of thrust drop depends on propeller type (with hinged or rigid blades attachments), its aerodynamic and geometric characteristics. These dependencies should be clarified in future works.

In further considerations, let us rely on the fact that the thrust drop at the vortex ring state still exists. Therefore, the question arises whether the experimentally obtained thrust drop can cause helicopter unanticipated yaw rotation?

Let the helicopter be balanced with no crosswind. With crosswind appearance and its velocity, gradual increase in the direction opposite to the propeller throwback velocity first the TR thrust will increase up to the velocity $V = 4$ m/s. Then, it will decrease to a minimum at velocity $V = 6$ m/s but the thrust in these conditions will be almost equal to the initial thrust without crosswind blowing. After that with a further wind velocity increase, the propeller thrust will start to increase again. In these conditions, with no pilot control input helicopter will yaw to the direction opposite to the unanticipated yaw rotation (for European-made helicopters the unanticipated yaw happens to the left, for US-made helicopters—to the right; it is determined by MR rotation direction and its torque which is compensated by the TR). It should be mentioned that as can be seen in Fig. 3 the thrust drop is eliminated at β more than 40°.

If the helicopter is balanced with crosswind velocity $V = 4$ m/s then with wind velocity increase the thrust will decrease which can generally lead to the start of unanticipated yaw rotation. But it should be mentioned that in these conditions helicopter airframe aerodynamic yaw moment $M_{y\,\mathrm{ha}}$ will increase as well and it will prevent helicopter rotation (Fig. 5; also shown: $M_{y\,\mathrm{rMR}}$—main rotor torque, $M_{y\,\mathrm{TR}}$—TR thrust moment, T_{TR}—TR thrust).

It is also possible that tail rotor thrust decreases due to the decrease of TR rotation speed because of required power increase above available power [10]. However, when helicopter rotates on yaw from the position shown in Fig. 5 airflow blowing becomes side-blowing instead of axial blowing and vortex ring state eliminates when β exceeds 40°. The power required for TR rotation should significantly decrease and its rotation speed should be restored. However, this is a speculative conclusion that requires deeper study and additional research.

Attention should be also paid to the real unanticipated yaw rotation events when the helicopter angular yaw speed rapidly increases up to 100 °/s and more (in some cases, it reached 165 °/s [21]). Authors of this paper think that this can happen if only TR thrust decreases dramatically almost to zero and

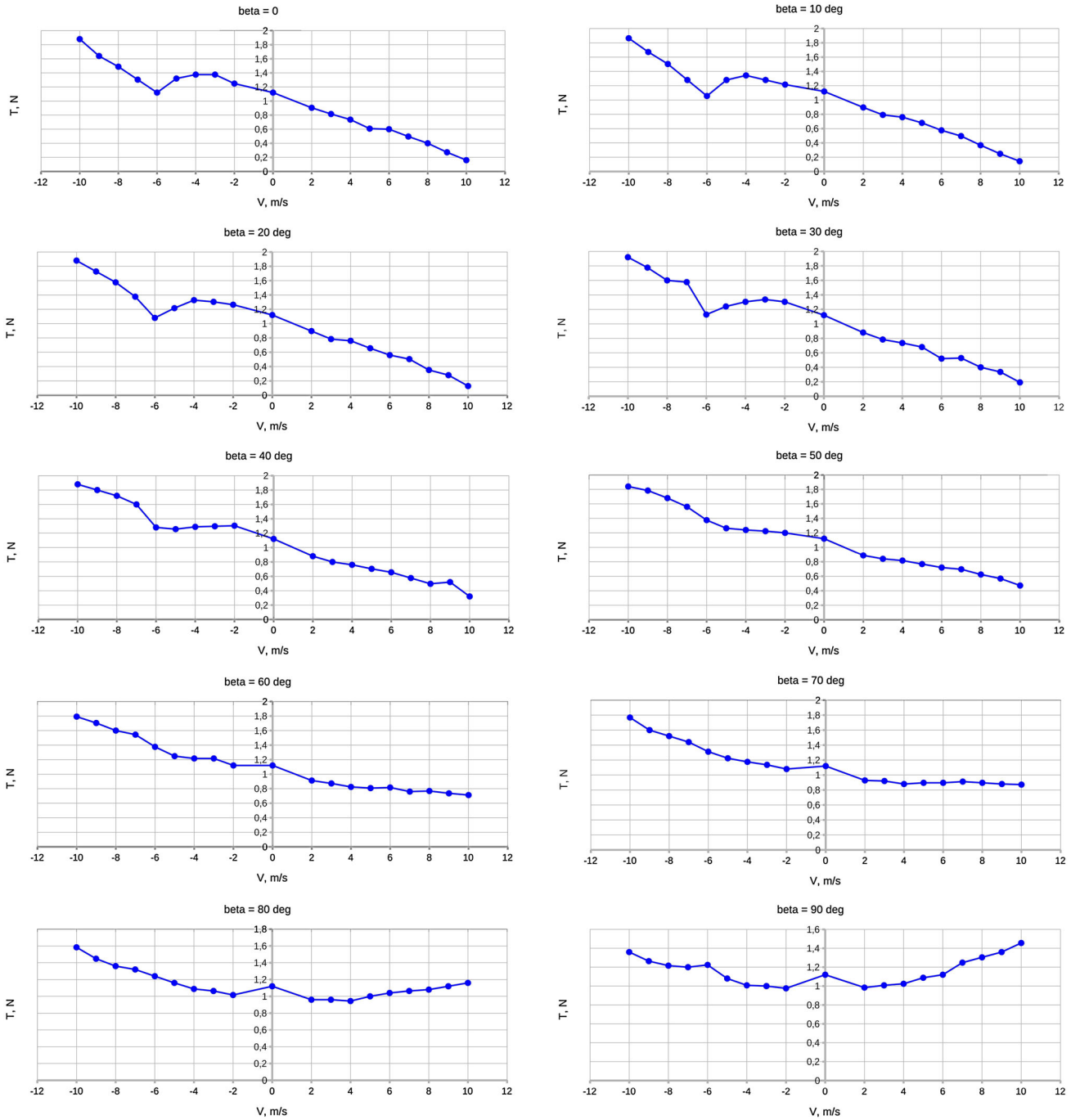


Fig. 3 Dependencies of propeller thrust on airflow velocities at different angles between propeller axis of rotation and airflow velocity vector

should be maintained at this value for a significant time for the helicopter to reach such high angular rotation speed. But as was mentioned above, the vortex ring state eliminates rather quickly and TR thrust is restored while transiting from its axial blowing to side-blowing.

If we return to Fig. 3 and pay attention to thrust dependencies at positive airflow velocities there is a monotonous decrease of propeller thrust while airflow velocity increases.

If the helicopter is balanced with no crosswind then with the occurrence of a side-wind gust with $V = 10$ m/s, the propeller thrust will drop by almost 80%. Also judging by experiments results shown in Fig. 3, this wind direction has a negative effect on the TR thrust in a wide range of angles $\beta = 0\ldots70$ degrees and so it has a comparatively longer negative effect on the TR thrust.

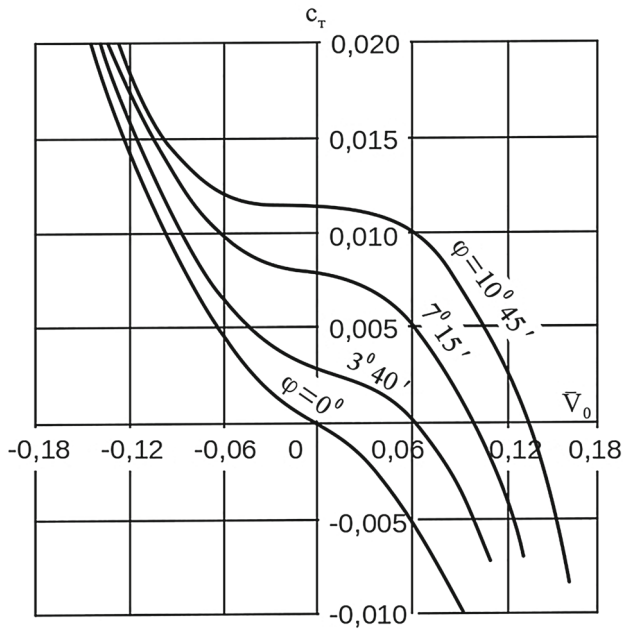


Fig. 4 Dependencies of propeller relative thrust on relative airflow velocity at different propeller pitch angles [22]

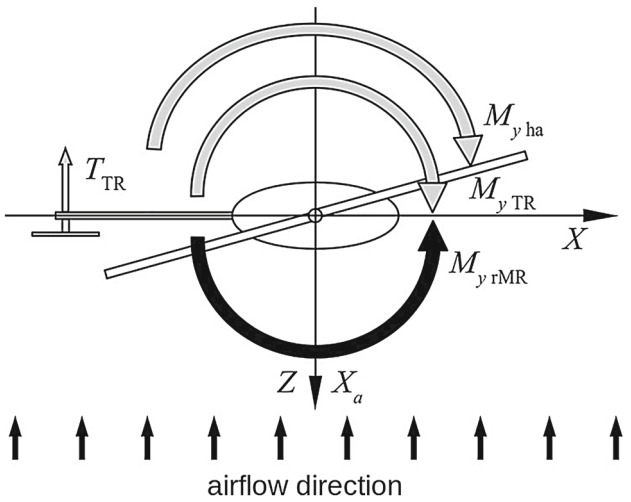


Fig. 5 Balance of moments affecting the helicopter while airflow blowing for increasing TR blades sections angles of attack

It should be also taken into account that in these conditions, helicopter airframe aerodynamic yaw moment $M_{y\,ha}$ contributes to the occurrence of unanticipated yaw rotation (Fig. 6).

These considerations suggest that helicopter wind blowing which decreases tail rotor blades angles of attack (against TR thrust vector) is more dangerous than blowing in the opposite direction which leads to the vortex ring state. In this regard, it is worth mentioning this paper authors experiments with a radio-controlled helicopter model in the wind tunnel which confirm this conclusion [23].

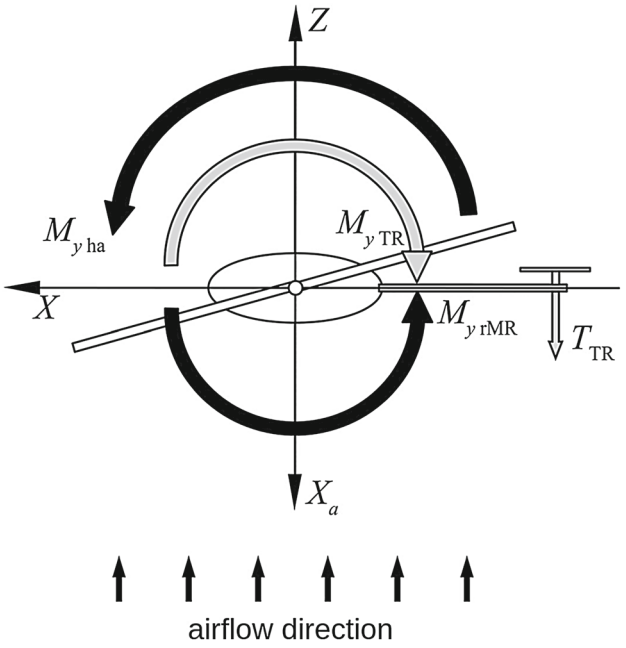


Fig. 6 Balance of moments affecting the helicopter while airflow blowing for decreasing TR blades sections angles of attack

4 Conclusion

Above presented experiments with a propeller in a wind tunnel show that when the propeller is blown for increasing blades sections angles of attack (in the direction of propeller thrust vector), the propeller thrust generally increases. But in the range of angles between propeller axis of rotation and airflow velocity vector from 0° to 40°, the vortex ring state occurs which causes a thrust drop at airflow velocities from about 4 to 7 m/s. Under certain conditions, this thrust drop can cause the unanticipated yaw rotation. However, according to the authors, this thrust drop is insignificant which nevertheless can increase due to the tail rotor rotation speed decrease on a real helicopter (because of the increase of required power due to the increase of blades sections angles of attack at the vortex ring state). Although this thrust drop should not be long-lasting, because while helicopter rotates on yaw the vortex ring state eliminates when angle between TR rotation axis and airflow velocity vector exceeds 40°.

When the propeller is blown for decreasing blades sections angles of attack (in the direction against propeller thrust vector) in the range of angles between propeller axis of rotation and airflow velocity vector from 0° to 70° there is a monotonous decrease of propeller thrust. While being axial blowing with airflow velocity $V = 10$ m/s the propeller thrust drops by 80% relative to the thrust without blowing, which may lead to the occurrence of unanticipated yaw rotation. Besides this, with this blowing direction, helicopter airframe

aerodynamic yaw moment contributes to the occurrence of unanticipated yaw rotation.

In this regard, helicopter wind blowing which decreases tail rotor blades angles of attack (against TR thrust vector) can be more dangerous than blowing in the opposite direction which leads to the vortex ring state. However, this paper authors believe that it is too soon to provide the final conclusions, because propeller used in the experiments was significantly different from the real helicopters tail rotors, interference with main rotor and many other aspects were not taken into account.

Author contributions All authors contributed to the study conception and design. All authors read and approved the final manuscript.

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Availability of data and materials Wind tunnel, mechanical scales, electric motor, propeller and measuring instruments are available at the Aerodynamics and Flight Dynamics Laboratory of Aerodynamics and Aircraft Structures Department of Moscow State Technical University of Civil Aviation.

Code availability No custom software was used. Graphs and drawings were designed using LibreOffice software.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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